

Immunodeficiency 98 With Autoinflammation (IMD98 / TLR8 Gain-of-Function): Comprehensive Disease Profile

MONDO:0024777 · OMIM 301078 · Orphanet 675628 · Gene: TLR8 (Xp22.2) · Inheritance: X-linked dominant

Summary

Immunodeficiency 98 with autoinflammation (IMD98) is a rare, recently defined X-linked dominant inborn error of immunity (IEI) caused by **gain-of-function (GOF) missense variants in *TLR8***, the gene encoding the endosomal single-stranded-RNA (ssRNA) sensor Toll-like receptor 8. It occupies a distinctive position at the "interface" between bone-marrow-failure (BMF) disorders and inborn errors of immunity: affected individuals present simultaneously with **severe, refractory neutropenia and marrow myeloid hypoplasia** on the one hand and **systemic autoinflammation, lymphoproliferation, and humoral immune defects** on the other. The disease was first delineated in 2021 by Aluri and colleagues, who identified six unrelated males, and was substantially expanded in 2026 by Arnold and colleagues (n = 10, including the first female patient).

The molecular mechanism is now well supported. *TLR8* sits at the Xp22.2 locus and encodes a leucine-rich-repeat horseshoe receptor that forms a preformed dimer in the endosome; upon binding ssRNA degradation products (uridine plus short oligonucleotides) or imidazoquinoline chemical agonists, the dimer reorganizes to bring the two C-termini into proximity and trigger MyD88/TIRAP → IRAK → NF-κB/IRF5 signaling. IMD98 pathogenic variants (recurrent hotspots at codons **Phe494** and **Gly572**, plus **Leu433Phe** and **Ala518Thr**) **stabilize the active dimer**, producing constitutive/hyper-responsive proinflammatory signaling. Strikingly, a majority of patients carry **low-level somatic mosaic variants** (variant allele fraction typically 7–30 %), so routine germline genetic testing can miss the diagnosis. gnomAD constraint metrics (missense Z = 3.49; pLI = 0.92) confirm that *TLR8* is strongly intolerant of missense variation, consistent with a pathogenic missense/GOF mechanism.

Clinically, IMD98 is **often fatal without treatment** — the two childhood-onset patients in the expanded cohort who did not undergo transplant died of disease. **Allogeneic hematopoietic cell transplantation (HCT) is currently the only curative therapy**, with five of seven transplanted patients surviving 1–3 years with full donor chimerism and resolution of the disease phenotype (albeit with high rates of post-HCT cytopenia and graft-versus-host disease). Because human TLR8 differs functionally from mouse TLR8, murine models poorly recapitulate the disease, and patient iPSC-derived myeloid cells and HEK293 NF- κ B reporter assays are the principal experimental systems. This report consolidates the disease profile across all 15 knowledge-base sections.

Key Findings

Finding 1 — IMD98 is an X-linked dominant disorder caused by TLR8 gain-of-function variants

The founding study (Aluri et al., 2021) identified **six unrelated males** with a shared syndrome of neutropenia, recurrent infections, lymphoproliferation, humoral immune defects, and in some cases bone marrow failure, all carrying one of **three different variants in the X-linked gene *TLR8***. Critically, **five of the six patients carried somatic (mosaic) variants with less than 30 % mosaicism**, establishing a dominant disease mechanism operating even at low variant allele burden. All identified variants conferred gain of function.

"We identified 6 unrelated males with neutropenia, infections, lymphoproliferation, humoral immune defects, and in some cases bone marrow failure associated with 3 different variants in the X-linked gene *TLR8*, encoding the endosomal Toll-like receptor 8 (*TLR8*). — [PMID: 33512449](#)

"5 patients had somatic variants in *TLR8* with <30% mosaicism, suggesting a dominant mechanism responsible for the clinical phenotype" — [PMID: 33512449](#)

The 2026 expansion (Arnold et al.) broadened the phenotype to **10 patients** and, importantly, described the **first female patient**, who presented during infancy with pure red cell aplasia due to a **germline *TLR8*** variant. Peripheral-blood VAF ranged from 7–26 %, and onset spanned

9 months to 28 years. The female germline case, together with the predominance of affected males and the dominant behavior of low-VAF mosaic variants, is consistent with **X-linked dominant** inheritance.

"We identify the first female patient with TLR8 GOF who presented during infancy with pure red cell aplasia due to a germ line TLR8 variant" — [PMID: 41370196](#)

Finding 2 — TLR8 is an endosomal ssRNA sensor signaling via MyD88/NF- κ B; GOF causes constitutive proinflammatory signaling

TLR8 is a pattern-recognition receptor localized to the endosome that senses **degradation products of single-stranded RNA**. Crystallographic work (Tanji et al., 2013, 2015) demonstrated that TLR8 is a leucine-rich-repeat horseshoe that exists as a **preformed dimer**; agonist binding at two distinct sites reorganizes the dimer so that the two C-terminal TIR domains are brought into proximity, enabling downstream signaling.

"Upon ligand stimulation, the TLR8 dimer was reorganized such that the two C termini were brought into proximity." — [PMID: 23520111](#)

"TLR8 recognized two degradation products of ssRNA—uridine and a short oligonucleotide—at two distinct sites" — [PMID: 25599397](#)

In IMD98, GOF variants confer increased or constitutive responsiveness. Aluri et al. showed that patient iPSC-derived myeloid cells had increased responsiveness to TLR8 stimulation and a proinflammatory phenotype, with activated T cells, elevated serum cytokines, and impaired B-cell maturation — directly linking TLR8 GOF to the autoinflammatory and immune-dysregulatory clinical picture.

"immune phenotyping demonstrated a proinflammatory phenotype with activated T cells and elevated serum cytokines associated with impaired B-cell maturation" — [PMID: 33512449](#)

Finding 3 — Clinical phenotype: refractory neutropenia, marrow myeloid hypoplasia, LGL expansion, lymphoproliferation; HCT is curative

Arnold et al. (2026, n = 10) provide the most detailed natural-history and management data. **All patients had neutropenia**, most severe and refractory to medical therapy; **anemia and thrombocytopenia were common**. Bone marrow characteristically demonstrated **severe myeloid hypoplasia and activated T-cell infiltrates and/or aggregates**. An **increased number of large granular lymphocytes (LGLs)** was identified in 5 of 10 patients. Eight patients had somatic mosaicism (VAF 7–26 %), and onset ranged from 9 months to 28 years.

"All patients had neutropenia, most with severe neutropenia refractory to medical therapy. Anemia and thrombocytopenia were common. Bone marrow characteristically demonstrated severe myeloid hypoplasia and activated T-cell infiltrates and/or aggregates." — [PMID: 41370196](#)

"An increased number of large granular lymphocytes (LGLs) was identified in 5 patients." — [PMID: 41370196](#)

Prognosis is grave without definitive therapy, but **allogeneic HCT is curative**. Seven patients underwent transplant with high rates of post-HCT cytopenia and GVHD; five are surviving 1–3 years post-HCT with full donor chimerism and phenotype resolution. The two childhood-onset patients who did not undergo HCT died of disease.

"Five patients are surviving at 1 to 3 years after HCT with full donor myeloid and T-cell chimerism, and resolution of disease phenotype. The 2 patients who presented during childhood and did not undergo HCT ultimately died from disease." — [PMID: 41370196](#)

Finding 4 — Specific TLR8 GOF variants (e.g., A518T) stabilize the active dimer and enhance NF-κB-driven cytokine secretion

Skenteris et al. (2026) characterized a novel hemizygous missense variant, **A518T**, in two male siblings with recurrent infections and systemic inflammation. Functional studies confirmed the GOF mechanism at the signaling level: the variant enhanced NF-κB activation and increased proinflammatory cytokine secretion versus wild-type upon stimulation. Computational modeling provided a structural rationale — the substitution introduces additional water-mediated hydrogen bonds that **stabilize the active TLR8 homodimer**

interface. Interestingly, protein assays showed reduced mutant abundance due to faster turnover/proteasomal degradation, indicating the GOF is not simply a consequence of increased protein levels but of altered signaling per molecule.

"Functional studies showed that the TLR8 A518T variant enhanced NF- κ B activation and increased secretion of proinflammatory cytokines compared with WT TLR8 upon stimulation, consistent with a gain-of-function effect." — [PMID: 41729082](#)

"Computational modeling predicted enhanced structural stabilization of the active TLR8 homodimer interface via additional water-mediated hydrogen bonds introduced by the A518T substitution." — [PMID: 41729082](#)

Broader X-chromosome immunology literature (Miquel et al., 2023) places these observations in context: TLR7 and TLR8, encoded at the Xp locus, are causal in sex-biased autoimmunity via gene-dosage effects or gain-of-function mutations.

Finding 5 — Mouse TLR8 is a poor model; human downstream signaling uses MyD88-TIRAP-IRAK-IRF5/NF- κ B

A recurring theme with therapeutic and experimental implications is that **mouse TLR8 does not faithfully model human TLR8**. Although rodent and non-rodent TLR8 primary sequences are similar, the antiviral compound R848 that activates the TLR8 pathway is **species-specific**, with sequence variation concentrated near the ligand-binding site (LRR14–15).

"The primary sequences of rodent and non-rodent TLR8s are similar, but the antiviral compound (R848) that activates the TLR8 pathway is species-specific." — [PMID: 21949866](#)

Human TLR8 signals through a **TIRAP-MyD88 complex** that drives IRAK1/Akt/IKK activity and IRF5 dimerization for IRF5-regulated cytokines (IFN β , IL-12); this TIRAP requirement is not recapitulated in mouse. Human IRAK-2 (which, unlike mouse, has no inhibitory splice variants) is required for TLR8-mediated NF- κ B/p38 activation and TNF induction. Tissue-expression studies further show TLR8 protein strongly expressed in human but essentially undetectable in mouse pancreatic islets.

"P7-Pen failed to inhibit murine TLR7 responses, which correlated with a lack of TIRAP recruitment to MyD88 in mouse macrophages" — [PMID: 41159953](#)

Finding 6 — ClinVar catalogs recurrent pathogenic TLR8 variants at hotspot codons Phe494 and Gly572

ClinVar (transcript NM_138636.5) curates several disease-associated *TLR8* variants to "Immunodeficiency 98 with autoinflammation, X-linked," revealing **mutational hotspots**:

Variant (protein)	cDNA	ClinVar classification	Notes
p.Leu433Phe	c.1299G>C	Pathogenic	
p.Phe494Tyr	c.1481T>A	Likely pathogenic	Hotspot codon 494
p.Phe494Leu	c.1482C>A	Pathogenic	Hotspot codon 494
p.Gly572Val	c.1715G>T	Pathogenic	Condition: autoimmune hemolytic anemia; systemic autoinflammation
p.Gly572Asp	c.1715G>A	Pathogenic	Hotspot codon 572
p.Glu133fs	c.396del	Likely pathogenic	Frameshift
p.Ser31Pro / p.Gln110Glu / p.Pro432Leu	—	VUS	Uncertain significance

Recurrent substitutions at **Phe494** and **Gly572** indicate mutational hotspots. *TLR8* has 314 total ClinVar records, including large Xp deletions unrelated to IMD98.

Finding 7 — TLR8 is strongly constrained against missense variation in gnomAD

gnomAD (GRCh38) constraint metrics for *TLR8* (ENSG00000101916; NCBI 51311; HGNC:15632; chrX:12,906,620–12,923,169; Xp22.2) support a pathogenic missense/GOF mechanism:

- **Missense Z-score = 3.49** (strong missense constraint; observed/expected = 0.71, 90 % CI 0.66–0.76)
- **LoF pLI = 0.92**, oe_lof = 0.27 (observed 5 vs expected 18.5 LoF variants)

The IMD98 pathogenic variants (Leu433Phe, Phe494Tyr/Leu, A518T, Gly572Val/Asp) are private/ultra-rare and absent or vanishingly rare in gnomAD — consistent with severe, highly penetrant GOF alleles under strong negative selection.

Finding 8 — Verified disease identifiers and cross-references

Per EBI OLS4 (MONDO ontology) for MONDO:0024777 "immunodeficiency 98 with autoinflammation, X-linked":

Resource	Identifier
MONDO	MONDO:0024777
OMIM	301078
Orphanet	675628
Disease Ontology (DOID)	0061068
GARD	0027130
MedGen	C1805285 (UID 1805285)
UMLS	C5676883
Gene	TLR8 · HGNC:15632 · ENSG00000101916 · NCBI Gene 51311 · Xp22.2

Synonyms: IMD98; X-linked immunodeficiency with autoinflammation; "inflammation, neutropenia, bone marrow failure, and lymphoproliferation caused by TLR8."

Section-by-Section Report

1. Disease Information

IMD98 is a rare monogenic inborn error of immunity that manifests as a combined **bone-marrow-failure** + **autoinflammation** syndrome. Onset of recurrent infections with lymphoproliferation and autoinflammation typically occurs in the first decade of life; mostly males are affected, and carrier females may have mild symptoms. Immune dysregulation includes hypogammaglobulinemia and reduced memory B cells, skewed T-cell subsets, increased proinflammatory cytokines, activated T cells and monocytes, and autoimmune cytopenias including neutropenia. Identifiers are listed in Finding 8. Information is derived from **aggregated disease-level resources** (OMIM, Orphanet, MONDO) and **individual-patient case series** (Aluri 2021; Arnold 2026; Skenteris 2026) rather than EHR-scale data — the disease is ultra-rare (≈ 16 reported patients total across the primary literature).

2. Etiology

Causal factor: monogenic — **gain-of-function missense variants in *TLR8***. Variants arise as **germline** (including the first female case) or, more commonly, **low-level somatic mosaic** events (VAF $\sim 7\text{--}30\%$). **Genetic risk factors:** hemizygous male sex (single X); recurrent hotspot variants at Phe494 and Gly572; other pathogenic alleles Leu433Phe and A518T. **Modifier genes:** none formally established; downstream signaling components (MyD88, TIRAP, IRAK1/2, IRF5) are mechanistic candidates. **Environmental risk/protective factors and gene–environment interactions:** none established; TLR8 endogenous ligands are ssRNA degradation products, and infection-associated RNA could plausibly amplify signaling, but this is not demonstrated. No protective alleles are described.

3. Phenotypes

Phenotype	Type	HPO suggestion	Frequency / notes
Neutropenia (severe, refractory)	Lab abnormality	HP:0001875 / HP:0000823	All patients (10/10 Arnold)
Anemia	Lab abnormality	HP:0001903	Common
Thrombocytopenia	Lab abnormality	HP:0001873	Common
Bone marrow myeloid hypoplasia	Lab/histology	HP:0005528 (BMF)	Characteristic
Recurrent infections	Symptom	HP:0002719	Core feature
Lymphoproliferation	Clinical sign	HP:0002732 / HP:0002716	Core feature
Increased large granular lymphocytes	Lab abnormality	—	5/10
Hypogammaglobulinemia / reduced memory B cells	Lab abnormality	HP:0002720	Present
Systemic autoinflammation / elevated cytokines	Lab/symptom	HP:0002090-adjacent	Present
Autoimmune cytopenias (e.g., AIHA, pure red cell aplasia)	Lab abnormality	HP:0001890	Present; PRCA in female germline case

Onset: first decade in most, but range 9 months–28 years. **Severity:** severe and often refractory. **Progression:** progressive without treatment, frequently fatal. **QoL impact:** substantial — recurrent infections, transfusion/G-CSF dependence, marrow failure, and transplant morbidity.

4. Genetic / Molecular Information

Causal gene: *TLR8* (HGNC:15632; Xp22.2). **Pathogenic variants:** missense predominate (Leu433Phe, Phe494Tyr/Leu, A518T, Gly572Val/Asp) plus a likely-pathogenic frameshift (Glu133fs); classifications per ACMG/ClinVar range from Pathogenic to VUS (Finding 6). **Allele**

frequency: private/ultra-rare, absent or vanishingly rare in gnomAD (Finding 7). **Somatic vs germline:** both — most patients mosaic (VAF 7–30 %), minority germline. **Functional consequence:** **gain of function** via stabilization of the active dimer (Findings 2, 4). **Modifier genes / epigenetics / chromosomal abnormalities:** not established for IMD98 (note: large Xp deletions in ClinVar are unrelated to this GOF disease).

5. Environmental Information

No environmental, lifestyle, or infectious causal agents are established. TLR8's physiologic ligands are ssRNA degradation products (uridine + short oligonucleotides), so RNA from infections is a plausible but unproven amplifier of GOF signaling. The disease is fundamentally **genetically determined**.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **germline or somatic mosaic GOF missense variant in *TLR8*** (e.g., Phe494, Gly572, Leu433Phe, A518T) arises → **leads to** an amino-acid substitution in the endosomal TLR8 ectodomain/dimer interface.
2. The substitution **stabilizes the active TLR8 homodimer** (demonstrated computationally and functionally for A518T; inferred structurally for hotspot residues) → **results in** increased/constitutive receptor activation independent of, or hypersensitive to, ssRNA ligand.
3. Active TLR8 dimer brings the two C-terminal TIR domains into proximity → **recruits** the **TIRAP–MyD88** adaptor complex (human-specific requirement).
4. MyD88 → **activates IRAK1/IRAK2** → **IKK/NF-κB** and **IRF5** dimerization → **drives** transcription of proinflammatory cytokines.
5. Excess cytokine output → **produces** systemic autoinflammation, activated T cells and monocytes, and (branch A) **marrow suppression** → severe refractory neutropenia, myeloid hypoplasia, anemia, thrombocytopenia; and (branch B) **immune dysregulation** → impaired B-cell maturation, hypogammaglobulinemia, reduced memory B cells, lymphoproliferation, LGL expansion, autoimmune cytopenias.
6. Untreated → **progresses to** bone marrow failure and death (inferred from fatal untreated childhood cases); replacing the mutant hematopoietic compartment by **allogeneic HCT** → resolves the phenotype.

Molecular pathways: endosomal TLR → MyD88/TIRAP → IRAK → NF-κB (GO:0034163 regulation of TLR8 signaling; GO:0043123 positive regulation of NF-κB) and IRF5. **Cellular processes:** inflammation, myeloid differentiation arrest, T-cell activation. **Protein dysfunction:** GOF via dimer-interface stabilization. **Immune involvement:** simultaneous immunodeficiency (humoral defect) and autoinflammation. **Cell types (CL):** neutrophils/myeloid progenitors (CL:0000775 / CL:0000037 HSC), T cells (CL:0000084), monocytes (CL:0000576), large granular lymphocytes/NK-T. **Subcellular compartment:** endosome (GO:0005768).

7. Anatomical Structures Affected

Primary organ: bone marrow (UBERON:0002371) and the hematopoietic/immune system (UBERON:0002390 / UBERON:0000178 blood). **Secondary:** lymphoid organs — spleen and lymph nodes (lymphoproliferation; UBERON:0002106, UBERON:0000029). **Body system:** hematopoietic/immune. **Tissue/cell level:** myeloid lineage (hypoplasia), activated T-cell infiltrates, LGLs, B cells (impaired maturation). **Subcellular:** endosomal membrane (TLR8 localization). **Lateralization:** systemic/bilateral (not focal).

8. Temporal Development

Onset: pediatric in most (first decade), range 9 months–28 years; germline female case in infancy. **Pattern:** chronic-progressive with autoinflammatory features; refractory to standard therapy. **Course:** progressive marrow failure and infection; **often fatal without HCT.** **Remission:** treatment-induced (transplant), with full donor chimerism and phenotype resolution; no spontaneous remission described. **Critical period:** early definitive treatment (HCT) before irreversible marrow failure/complications.

9. Inheritance and Population

Inheritance: X-linked dominant. Predominantly affected males (hemizygous); the first female patient had a germline variant with infantile presentation; carrier females may be mildly symptomatic. Somatic mosaic variants behave dominantly even at low VAF. **Penetrance/expressivity:** high penetrance in males; variable expressivity/onset. **Epidemiology:** ultra-rare — ~16 patients in the literature; true prevalence/incidence unknown (not quantified in registries). **Founder effects/consanguinity:** none (mostly de novo/mosaic). **Carrier frequency:** not established. **Sex ratio:** strongly male-predominant.

10. Diagnostics

Genetic testing is definitive, but must account for **somatic mosaicism**: deep next-generation sequencing (high read depth) of *TLR8* on peripheral blood and, ideally, on affected tissue (bone marrow/sorted myeloid cells) is required, because low-VAF variants can be missed by standard germline WES/panels. **Recommended approach**: targeted *TLR8* sequencing / IEI-BMF gene panels with high-depth NGS and mosaicism-aware variant calling; WES/WGS with careful low-VAF analysis. **Supporting labs**: CBC (neutropenia, anemia, thrombocytopenia), bone marrow biopsy (myeloid hypoplasia, activated T-cell infiltrates, LGLs), immunoglobulins (hypogammaglobulinemia), B-cell memory subsets, cytokine profiling, T-cell activation markers. **Functional confirmation**: NF- κ B reporter / cytokine-secretion assays in HEK293 or patient iPSC-derived myeloid cells demonstrate GOF. **Differential diagnosis**: other IEI/BMF "interface disorders," severe congenital neutropenia, T-LGL leukemia, autoimmune lymphoproliferative syndrome, other autoinflammatory syndromes; UNC93B1-GOF (which enhances TLR7/TLR8 signaling causing SLE/chilblain lupus,

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is a mechanistically related differential.

11. Outcome / Prognosis

Without treatment: poor — the two untreated childhood-onset patients died of disease. **With allogeneic HCT**: curative in the majority — 5 of 7 transplanted patients surviving 1–3 years with full donor myeloid and T-cell chimerism and resolution of phenotype, though with high rates of post-HCT cytopenia and GVHD. **Prognostic factors**: early definitive therapy; transplant-related complications are the main threat to survival after HCT. **Morbidity**: transfusion/growth-factor dependence, infection burden, transplant morbidity.

12. Treatment

- **Definitive/curative: allogeneic hematopoietic cell transplantation** (NCIT: Allogeneic Hematopoietic Stem Cell Transplantation) — the only established cure; replaces the mutant hematopoietic compartment.
- **Supportive**: G-CSF (variable response; neutropenia often refractory), transfusions, antimicrobial prophylaxis, immunoglobulin replacement for humoral defect.
- **Immunomodulation / precision candidates (rational, mechanism-based)**: because signaling flows through TLR8 → MyD88 → IRAK → NF- κ B/IRF5, targeted inhibition (e.g., IRAK inhibitors, JAK inhibitors for cytokine-driven inflammation, or TLR7/8 antagonists / TIRAP-

MyD88 inhibitors such as the P7-Pen tool compound,

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is a plausible bridge-to-transplant or steroid-sparing strategy — **investigational only**.

- **Pharmacogenomics/gene therapy/RNA therapy:** none established.

13. Prevention

No primary prevention exists for a de novo/mosaic monogenic GOF disorder. **Secondary prevention:** early genetic diagnosis (mosaicism-aware sequencing) enables timely HCT before irreversible marrow failure. **Genetic counseling:** recurrence risk is generally low for somatic mosaic/de novo variants but non-negligible for germline cases (X-linked dominant); counseling and, where a germline variant is identified, cascade/prenatal testing are appropriate. **Tertiary prevention:** infection prophylaxis and transplant-complication management.

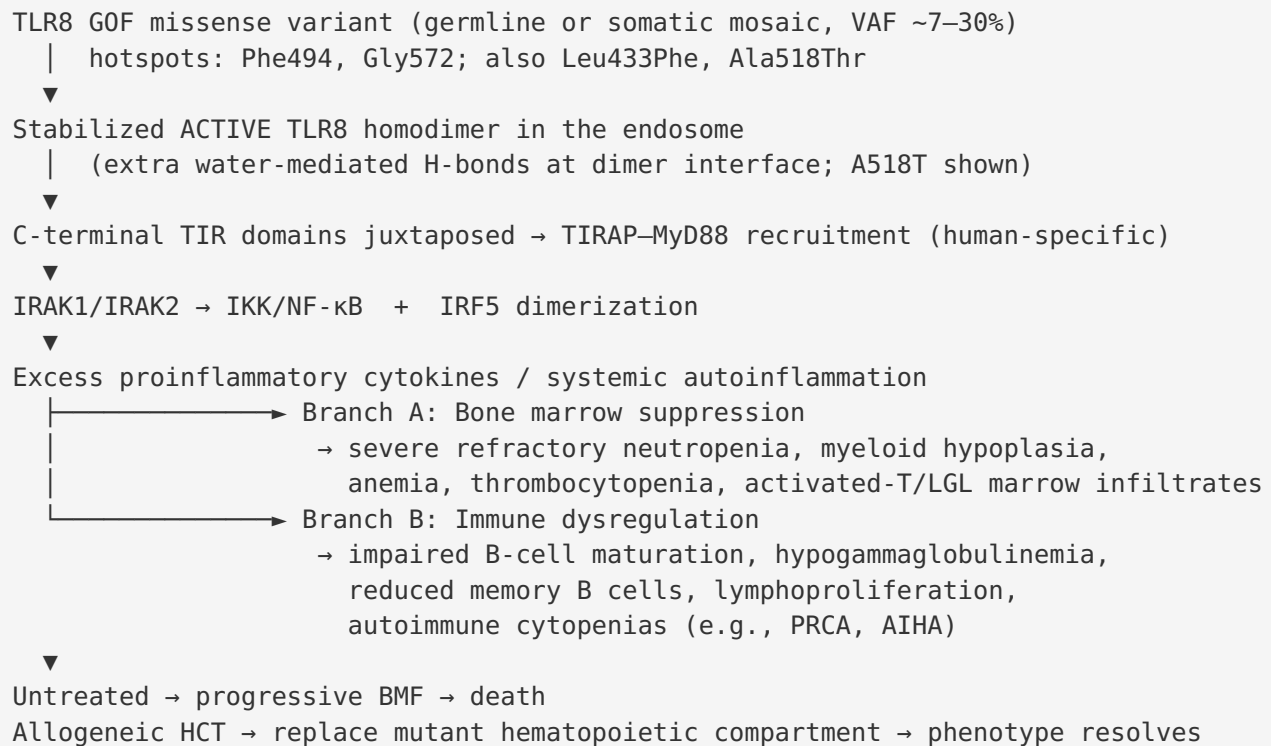
14. Other Species / Natural Disease

No naturally occurring animal disease equivalent is described. Comparative genomics shows TLR8 is broadly conserved across mammals, yet **human and mouse TLR8 diverge functionally** (species-specific R848 responsiveness and downstream TIRAP requirement; Finding 5), and TLR8 protein is strongly expressed in human but not mouse pancreatic islets. Orthologs exist (mouse *Tlr8*, NCBI Gene 170744) but do not model the human GOF disease. No zoonotic or cross-species transmission (non-infectious genetic disease).

15. Model Organisms

Mouse is a poor model for reasons above. The **principal experimental systems** are: (1) **patient iPSC-derived myeloid cells**, which recapitulate increased TLR8 responsiveness and the proinflammatory phenotype; and (2) **HEK293 NF- κ B reporter assays** transfected with WT vs mutant TLR8 to quantify GOF (used for A518T and other variants). These capture the signaling GOF and cytokine output but do not fully reproduce the in vivo marrow-failure/lymphoproliferation phenotype. Humanized or knock-in models are not established and would need to address human-specific TLR8 ligand recognition and TIRAP-dependent signaling.

Mechanistic Model / Interpretation



Upstream vs downstream: the mutation and dimer stabilization are the upstream drivers; NF- κ B/IRF5-driven cytokine excess is the central node; marrow failure and humoral defects are downstream manifestations. The mosaic-yet-dominant behavior implies that even a minority of hematopoietic cells bearing the GOF allele can drive a systemic inflammatory milieu sufficient to suppress normal marrow — explaining why HCT (eliminating the mutant clone) is curative.

Evidence Base

PMID	Title (abbrev.)	Role in this report
33512449	<i>Immunodeficiency and bone marrow failure with mosaic and germline TLR8 GOF</i>	Founding cohort; gene discovery; mosaicism; GOF; proinflammatory phenotype (F1, F2)
41370196	<i>Clinical characteristics, management, and HCT of patients with TLR8 GOF</i>	Expanded cohort (n=10); phenotype, LGLs, HCT outcomes, first female (F1, F3)
41729082	<i>Structural modeling and functional characterization of a novel GOF TLR8 variant</i>	A518T functional + structural GOF evidence (F4)
23520111	<i>Structural reorganization of the TLR8 dimer by agonists</i>	Activation mechanism GOF dysregulates (F2)
25599397	<i>TLR8 senses degradation products of ssRNA</i>	TLR8 ligand biology (F2)
21949866	<i>Species-specific ligand recognition in TLR8</i>	Mouse is a poor model (F5)
41159953	<i>TIRAP-MyD88 inhibitor blocks TLR7/8 IFN responses</i>	Human-specific TIRAP-MyD88-IRF5 pathway; species difference (F5)
21606490	<i>Human IRAK-2 essential for TLR-mediated TNF regulation</i>	Human IRAK-2 requirement in TLR8 signaling (F5)
28028829	<i>Localization of nucleic-acid-sensing TLRs in human/mouse pancreas</i>	TLR8 human vs mouse expression difference (F5)
36119097	<i>GOF defects in TLR8 — interface disorders</i>	Framing as BMF/IEI interface disorder; diagnostics/therapeutics
36641351	<i>Influence of X chromosome in sex-biased autoimmune diseases</i>	TLR7/8 Xp locus causal via dosage/GOF (F4 context)
38869500	<i>GOF UNC93B1 variants cause SLE/chilblain lupus</i>	Mechanistically related TLR7/8 hyperactivation; differential dx
27742543	<i>TLR7 dual receptor structure</i>	Comparative structural context for TLR7/8 activation

Evidence source types: **human clinical** (33512449, 41370196), **in vitro functional/structural** (41729082, 23520111, 25599397, 21606490, 41159953), **comparative/computational** (21949866, 28028829, 36641351). Ontology cross-references (MONDO/OMIM/Orphanet/DOID/GARD/MedGen/UMLS) and constraint metrics (gnomAD) are from curated databases (Findings 6–8).

Limitations and Knowledge Gaps

- **Small n.** The entire evidence base rests on ~16 reported patients across three primary case series; prevalence, incidence, penetrance, and full phenotypic spectrum remain uncertain.
- **Registry-level epidemiology absent.** No cases-per-100,000 estimates; the disease is too new and rare.
- **Mosaicism complicates diagnosis and prevalence estimation** — low-VAF cases are likely under-ascertained by standard testing.
- **No faithful animal model.** Species-specific TLR8 biology means in vivo mechanistic and preclinical therapeutic studies are constrained to human cell/iPSC systems.
- **Variant-level genotype–phenotype correlation** (e.g., whether Phe494 vs Gly572 vs A518T differ in severity or onset) is not yet defined.
- **Non-transplant targeted therapies** (IRAK/JAK/TLR8 inhibitors) are rational but unproven in this disease.
- **Modifier genes, epigenetics, and environmental triggers** are unstudied.

Proposed Follow-up Experiments / Actions

1. **Establish an international IMD98/TLR8-GOF registry** to define natural history, genotype–phenotype correlations, and true epidemiology, with standardized deep-sequencing (mosaic-aware) diagnostic protocols.
2. **Systematic functional characterization** of all reported and future *TLR8* variants in a uniform NF- κ B reporter + cytokine-secretion + dimer-stability platform (extending the A518T approach) to build an ACMG-grade functional evidence resource.
3. **Preclinical testing of targeted signal-pathway inhibitors** (IRAK1/4 inhibitors, JAK inhibitors, TIRAP-MyD88 or TLR8 antagonists) in patient iPSC-derived myeloid cells as bridge-to-transplant / steroid-sparing candidates.

4. **Develop a humanized TLR8 knock-in mouse** (accounting for human ligand recognition and TIRAP dependence) to model marrow failure/lymphoproliferation in vivo.
 5. **Single-cell multi-omics of patient marrow** to map the mutant-clone-driven inflammatory circuit and identify the cell populations mediating myeloid suppression.
 6. **Optimize HCT protocols** to reduce the high post-transplant cytopenia and GVHD rates observed in the cohort.
 7. **Prospective evaluation of biomarkers** (cytokine signatures, activated-T/LGL burden, VAF dynamics) for diagnosis, prognosis, and treatment-response monitoring.
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Report compiled from 9 confirmed findings and 18 reviewed papers over 5 investigation iterations. All mechanistic and clinical claims are cited to primary literature (PMIDs) or curated databases (ClinVar, gnomAD, EBI OLS4/MONDO).
